Data Mining Introduction

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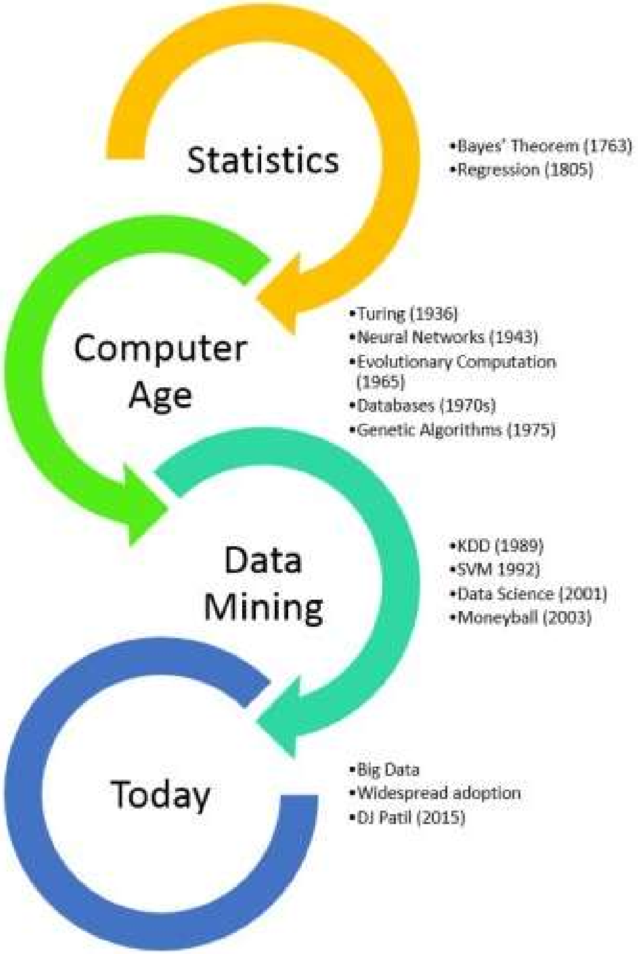


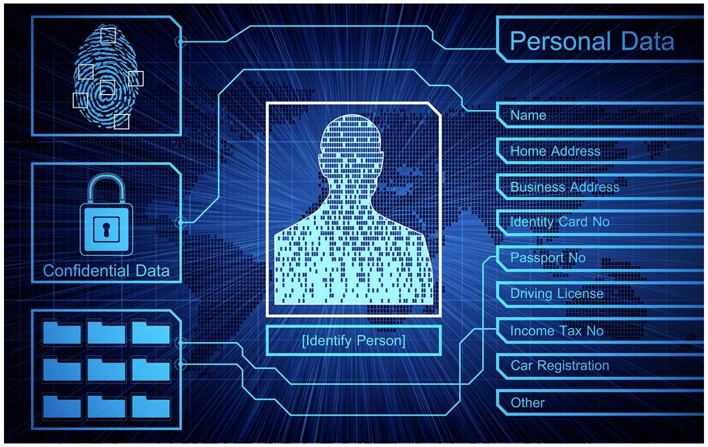
Evolutionary Step	**Business Question**	**Enabling Technologies**	**Product Providers**	**Characteristics**
Data Collection (1960s)	"What was my total revenue in the last five years?"	Computers, tapes, disks	IBM, CDC	Retrospective, static data delivery
Data Access (1980s)	"What were unit sales in New England last March?"	Relational databases (RDBMS), Structured Query Language (SQL), ODBC	Oracle, Sybase, Informix, IBM, Microsoft	Retrospective, dynamic data delivery at record level
Data Warehousing & Decision Support (1990s)	"What were unit sales in New England last March? Drill down to Boston."	On-line analytic processing (OLAP), multidimensional databases, data warehouses	Pilot, Comshare, Arbor, Cognos, Microstrategy	Retrospective, dynamic data delivery at multiple levels
Data Mining (Emerging Today)	"What's likely to happen to Boston unit sales next month? Why?"	Advanced algorithms, multiprocessor computers, massive databases	Pilot, Lockheed, IBM, SGI, numerous startups (nascent industry)	Prospective, proactive information delivery

Syllabus

- Lecture 1 : Data Preprocessing
- Lecture 2 : Explanatory Data Analysis
- Lecture 3 : Feature Engineering (Feature Importance and Selection)
- Lecture 4 : Association Rule Learning
- Lecture 5 : Unsupervised Clustering
- Lecture 6 : Unsupervised Clustering (cont.)
- Lecture 7 : Anomaly and Outlier Detection
- Lecture 8 : Regression and Classification Learning
- Lecture 9 : Recommendation Learning
- Lecture 10 : Final Project Requirement

Introduction

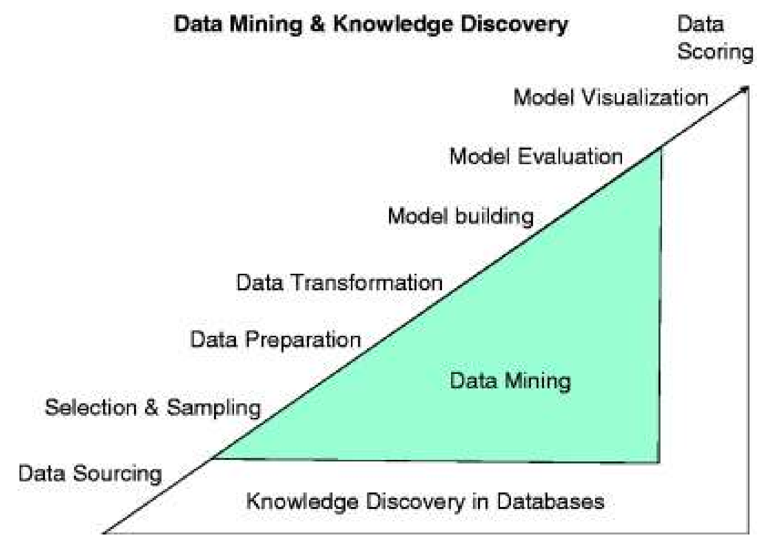
- By the early 1990`s, data mining was recognized as a sub-process or a step within a larger process called Knowledge Discovery in Databases (KDD)
- The most commonly used definition of KDD is "The nontrivial process of identifying valid, novel, potentially useful, and ultimately understandable patterns in data" (Fayyad, 1996).



Introduction

The sub-processes that form part of the KDD process are:

- Understanding of the application and identifying the goal of the KDD process
- Creating a target data set
- Data cleaning and pre-processing
- Matching the goals of the KDD process (step 1) to a particular data-mining method.
- Research analysis and hypothesis selection
- Data mining: Searching for patterns of interest in a particular form , including classification rules, regression, and clustering
- Interpreting mined patterns
- Acting on the discovered analysis



Data Mining Example (Marketing)

- **Marketing**. Data mining is used to explore increasingly large databases and to improve market segmentation. By analysing the relationships between parameters such as **customer age, gender, tastes**, etc., it is possible to guess their behaviour in order to direct personalised loyalty campaigns.
- Data mining in marketing **also predicts which users are likely to unsubscribe from a service,** what interests them based on their searches, or what a mailing list should include to achieve a higher response rate.

Data Mining Example (Retail)

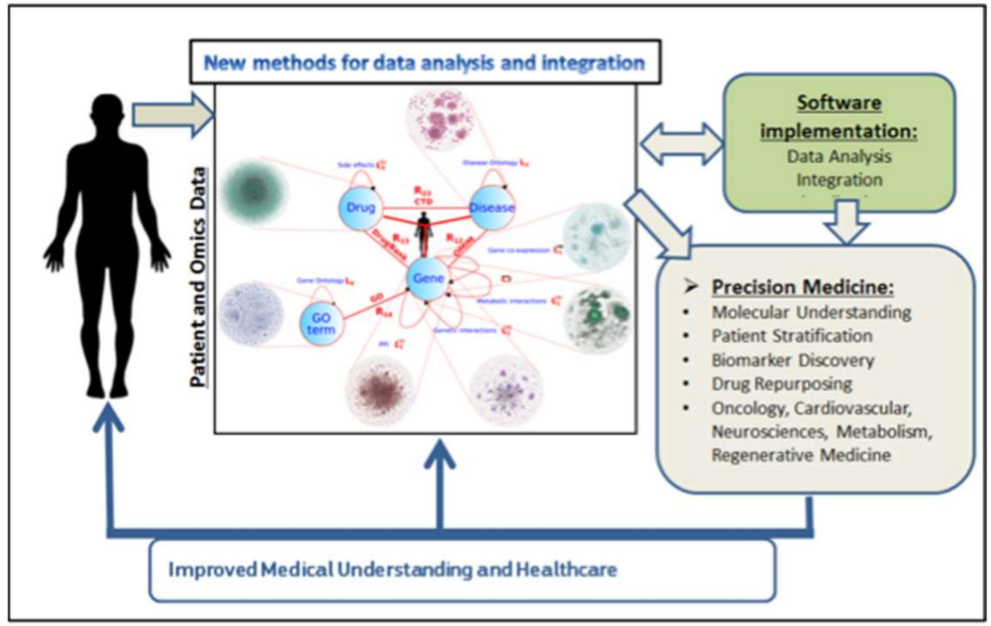
- **Retail**. Supermarkets, for example, use joint purchasing patterns to identify product associations and decide how to place them in the aisles and on the shelves.
- **Data mining also detects which offers are most valued by customers** or increase sales at the checkout queue.

Data Mining Example (Banking)

- **Banking**. Banks use data mining to better understand market risks. It is commonly applied to credit ratings and to intelligent anti-fraud systems to analyse transactions, card transactions, purchasing patterns and customer financial data.
- Data mining also **allows banks to learn more about our online preferences or habits** to optimise the return on their marketing campaigns, study the performance of sales channels or manage regulatory compliance obligations.

Data Mining Example (Medicine)

- **Medicine**. Data mining enables more accurate diagnostics. Having all of the patient's information, such as medical records, physical examinations, and treatment patterns, allows more effective treatments to be prescribed.
- It also **enables more effective, efficient and cost-effective management of health resources** by identifying risks, predicting illnesses in certain segments of the population or forecasting the length of hospital admission. Detecting fraud and irregularities, and strengthening ties with patients with an enhanced knowledge of their needs are also advantages of using data mining in medicine.



Data Mining Example
(Television and radio and Website)

- **Television and radio and Website**. There are networks that apply real time data mining to measure their online television (IPTV) and radio audiences. These systems collect and analyse, on the fly, anonymous information from channel views, broadcasts and programming.
- **Data mining allows networks to make personalised recommendations** to radio listeners and TV viewers, as well as get to know their interests and activities in real time and better understand their behaviour. Networks also gain valuable knowledge for their advertisers, who use this data to target their potential customers more accurately.

Summary about Data Mining

What is Data Mining?

Data mining is the process of sorting out the data to find something worthwhile. If being exact, mining is what kick-starts the principle "work smarter not harder."

Purpose of Data Mining

There are many purposes data mining can be used for. The data can be used for:

- **detecting** trends;
- **predicting** various outcomes;
- **modeling** target audience;
- **gathering** information about the product/service use;

Summary about Data Mining

Supervised machine learning algorithms are used for sorting out structured data:

- Classification is used to generalize known patterns. This is then applied to the new information (for example, to classify email letter as spam);
- Regression is used to predict certain values (usually prices, temperatures or rates);
- Normalization is used to flatten the independent variables of data sets and restructure data into the more cohesive form.

- **Unsupervised machine learning algorithms** are used for exploration of unlabeled data:
  - Clustering is used to detect distinct patterns (AKA groups AKA structures
  - Association rule learning is used to identify the relationship between the variables of the data set. For example, what kind of actions are performed most frequently;
  - Summarization is used for visualization and reporting purposes;

Summary about Data Mining

There is a critical challenge in handling all this data effectively and the challenge itself is threefold:

1. **Segmenting data** – recognizing important elements;
2. **Filtering the noise** – leaving out the noise;
3. **Activating data** – integrating gathered information into the business operation;

The evaluation of data mining applications

The main focus of data mining was tabular data; however with the evolving technology and different needs new sources were formed to be mined!

- *Text Mining:* Still a popular data mining activity, it categorizes or clusters large document collections such as news articles or web pages. Another application is opinion mining where the techniques are applied to obtain useful information from the questionnaire style data.
- *Image Mining*: In image mining, mining techniques are applied to images (2D and 3D)
- *Graph Mining:* It is formed from frequent pattern mining, which is focused on frequently occurring sub-graphs. A popular extension of graph mining is social network mining.

JOB Description

ManTech 4.0 ★

Data Scientist

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Duties, Tasks & Responsibilities

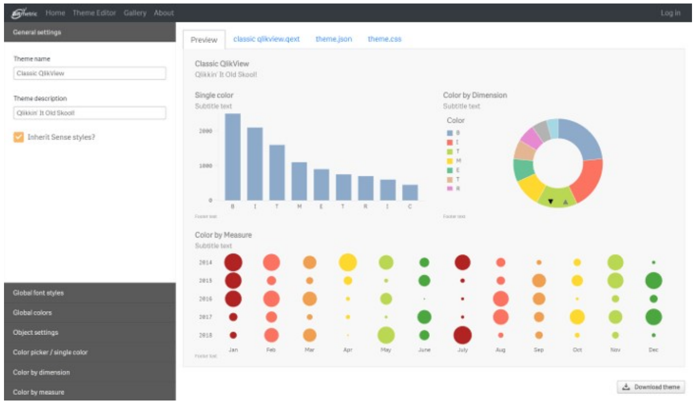
- Working with large, complex, and disparate data sets
- Designing and implementing innovative ways to analyze and exploit the Sponsors data holdings
- Researching and reporting on a wide variety of Sponsor inquiries
- Raising proactive inquiries to the Sponsor based on observations and proposed data analysis/exploitation
- Solving difficult, non-routine problems by applying advanced analytical methodologies, and improving analytic methodologies
- Developing custom searches
- Communicating and coordinating with internal and external partners as needed

Top Data Mining Tools

Qlik is Data mining and visualization tool. It also offers dashboards and Supports multiple data sources and file types.

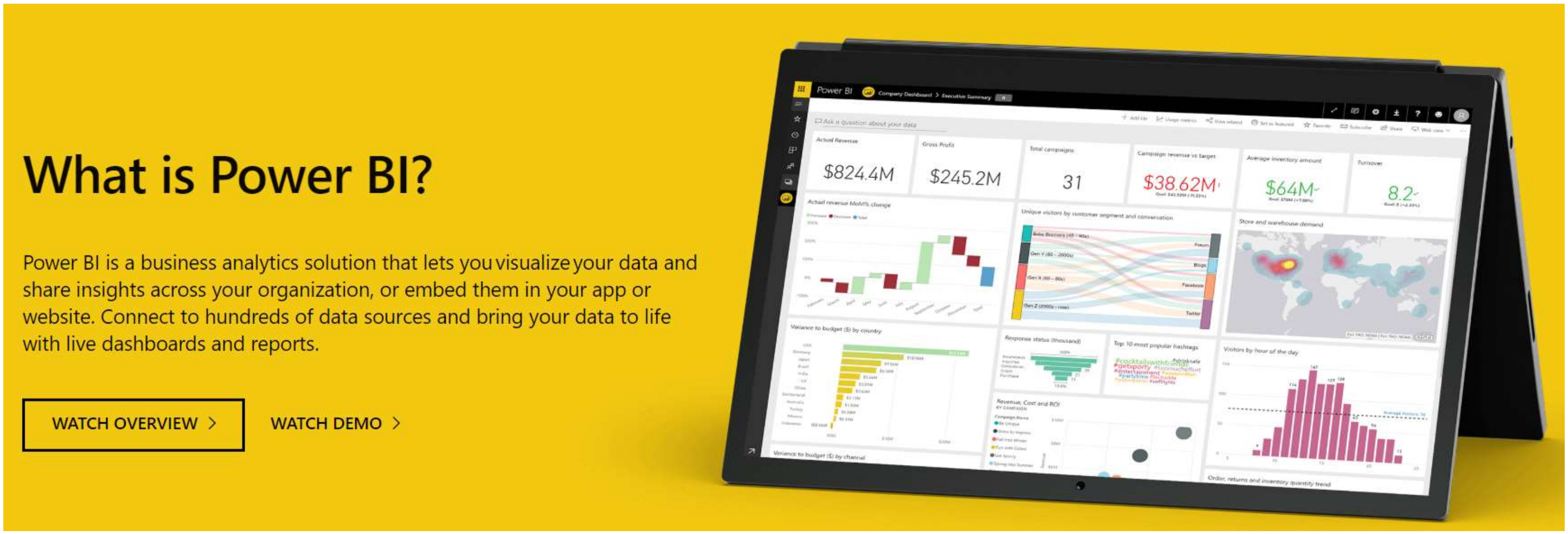
Features:

- Drag-and-drop interfaces to create flexible, interactive data visualizations
- Instantly respond to interactions and changes.
- Supports multiple data sources and file types
- It allows easy security for data and content across all devices.
- It allows you to share relevant analyses, including apps and stories, using a centralized hub.



Download link: https://www.qlik.com/us/products/qlik-sense

Top Data Mining Tools



Top Data Mining Tools

Tableau is business intelligence software that helps people see and understand their data.

Fast Analytics

Connect and visualize your data in minutes. Tableau is 10 to 100x faster than existing solutions.

Ease of Use

Anyone can analyze data with intuitive drag & drop products. No programming, just insight.

Big Data, Any Data

From spreadsheets to databases to Hadoop to cloud services, explore any data.

Smart Dashboards

Combine multiple views of data to get richer insight. Best practices of data visualization are baked right in.

Update Automatically

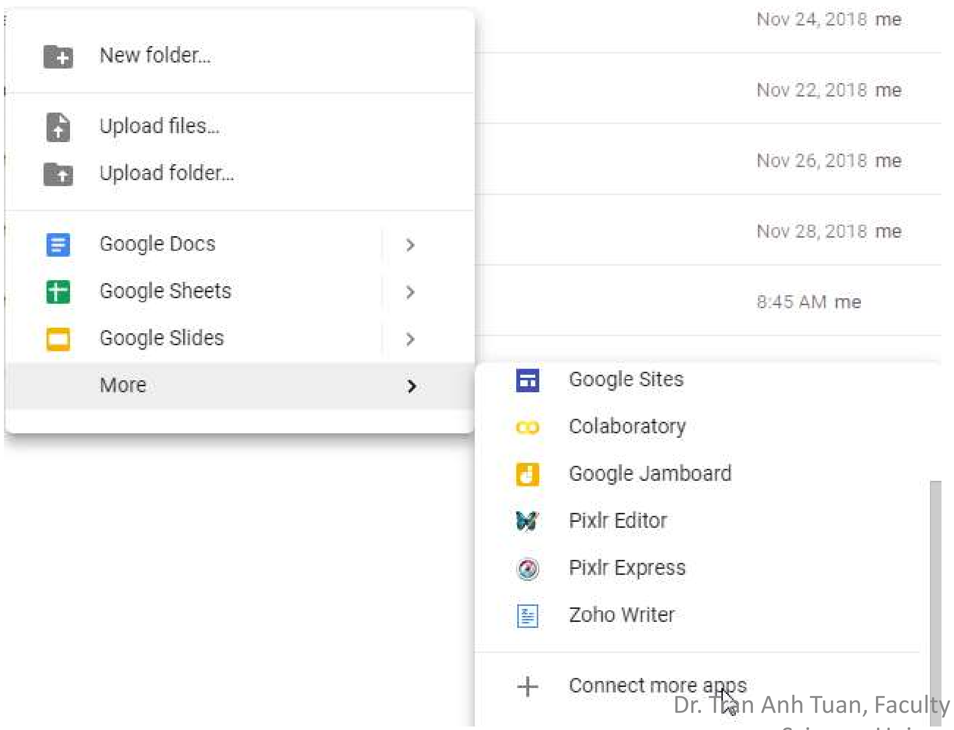
Get the freshest data with a live connection to your data or get automatic updates on a schedule you define.

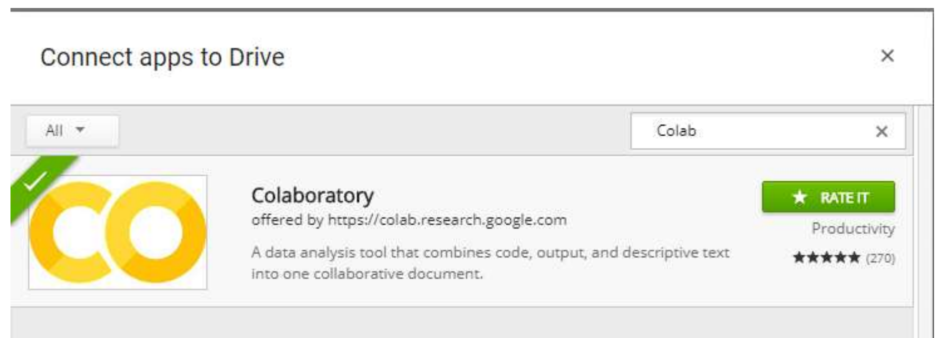
Share in Seconds

Publish a dashboard with a few clicks to share it live on the web and on mobile devices.

Data Mining With Python Code

- Colabs





Data Mining With Python Code

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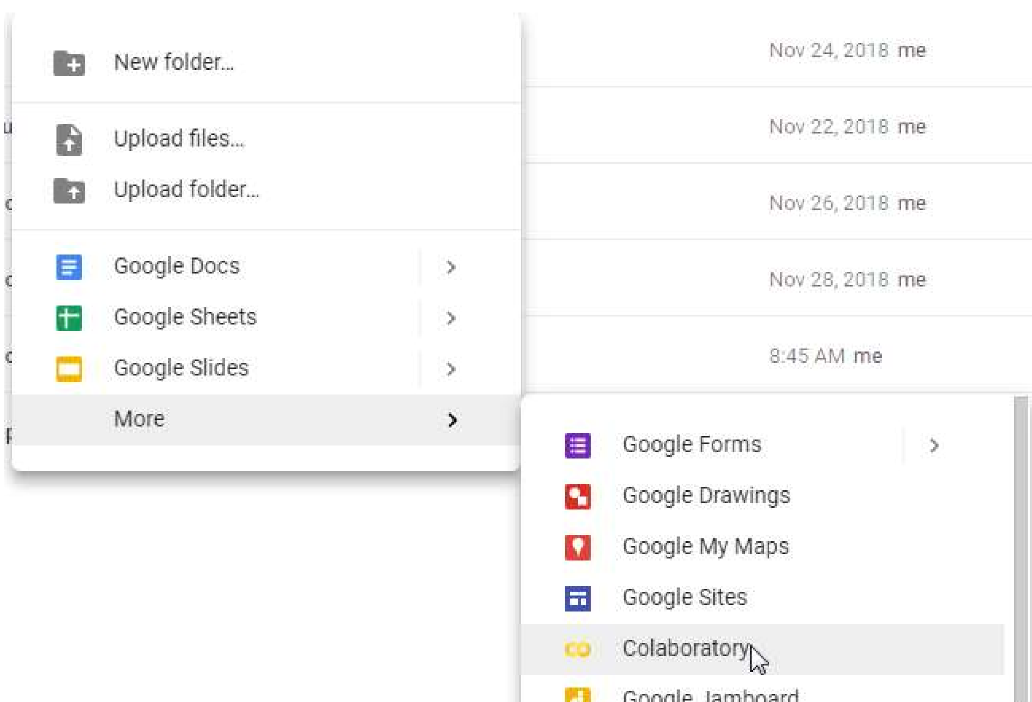

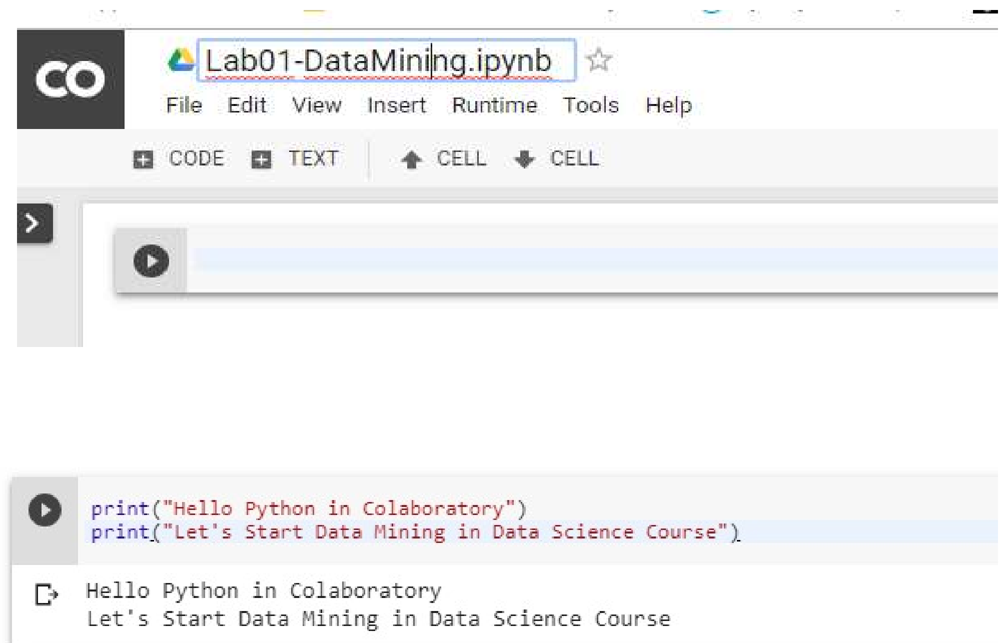

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